
(a) Uniform
(b) Band-restricted
(c) Filtered s11
(d) FE search
(e) Unconstrained
x3
x2
x1
t = 0.30
reference
t ∈ [0.20, 0.45]
KUBC +1.7% PBC −3.7%
t ∈ [0.12, 0.55]
KUBC +5.5% PBC −6.0%
t ∈ [0.12, 0.55]
KUBC +6.3% PBC −8.3%
t ∈ [0.12, 0.55]
KUBC −4.2% PBC −17.8%
0.12
0.20
0.30
0.40
0.55
local sheet thickness parameter t (wall 0.34–1.90 mm at a 15 mm cell)